

PLAUD AI

PHI-Safe Clinical Pipeline

A concept for privacy-first medical documentation built on Plaud recordings.

The Problem

Patient Data in Audio

Medical consultation recordings contain names, dates of birth, addresses, SSNs, and medical IDs. Sending raw audio or transcripts to cloud AI services is a compliance risk.

HIPAA Exposure

Most AI transcription and summarization tools route data through third-party cloud APIs. This creates liability under HIPAA for any healthcare organization using them.

No Off-the-Shelf Solution

There is no widely adopted tool that combines transcription, PHI detection, and AI summarization in a fully on-premise, HIPAA-friendly package.

What Was Built

1

Audio Input

Plaud recording or any
WAV/MP3 file

2

Transcription

OpenAI Whisper runs locally on
the device

3

PHI Detection

Microsoft Presidio flags names,
dates, SSNs, locations

4

Redaction

Each PHI token replaced inline:
<PERSON>, <DATE_TIME>

5

SOAP Note

Local Llama 3.2 model generates
clinical note from clean text

Zero PHI reaches the AI model at any point.

What is Available

Local Docker Deployment

Full pipeline runs in Docker. One command brings up the backend, Whisper, Presidio, and Ollama. Real-time transcription and redaction from any audio file.

Static Web Demo

Hosted at pavan-249.github.io/plaud-ai-demo. Shows pre-computed Whisper output from a real clinical consultation recording with PHI detected and redacted inline.

Live PHI Detection

Type or paste any transcript. PHI is highlighted and redacted in real time in the browser using a local regex engine, no server required.

Technology

Transcription	OpenAI Whisper	Runs fully on-device, no API key needed
PHI Detection	Microsoft Presidio + spaCy	Identifies 9+ entity types in real time
Redaction	Presidio Anonymizer	Inline replacement, original structure preserved
AI Summarization	Llama 3.2 via Ollama	Self-hosted, air-gap ready
Backend	FastAPI + Python 3.10	Lightweight, async REST API
Frontend	HTML / CSS / JS	Zero framework dependencies
Infrastructure	Docker Compose	One command, fully reproducible

Try It Now

pavan-249.github.io/plaud-ai-demo

github.com/Pavan-249/plaud-ai-demo